

Supplement to “Neural Networks for Partially Linear Quantile Regression”

Qixian Zhong

Department of Statistics and Data Science, School of Economics,
Wang Yanan Institute for Studies in Economics (WISE),
and MOE Key Laboratory of Econometrics, Xiamen University, Xiamen, China
and

Jane-Ling Wang

Department of Statistics, University of California, Davis, U.S.A.

This supplementary material contains the technical proofs and additional numerical studies for the results of the paper.

A Proof of Theorems

Proof of Theorem 3.1. Let $\hat{\beta} = (\hat{\theta}, \hat{m})$, $\beta_0 = (\theta_0, m_0)$ and $d(\beta_1, \beta_2) = [\mathbb{E}\{x^\top \theta_1 + m_1(Z) - x^\top \theta_2 - m_2(Z)\}^2]^{1/2}$, for any $\beta_1 = (\theta_1, m_1)$ and $\beta_2 = (\theta_2, m_2)$. We first show that

$$d(\hat{\beta}, \beta_0) \xrightarrow{p} 0, \text{ as } n \rightarrow \infty.$$

By Lemma 5 in Schmidt-Hieber (2020) and the fact $|\rho_\tau(u) - \rho_\tau(v)| \leq 2|u - v|$ for all $u, v \in \mathbb{R}$, we know that $\{\rho_\tau(Y - \theta^\top X - m(Z)) \mid \beta = (\theta, m) \in \mathbb{R}_A^p \times \mathcal{M}_A\}$ is *P-Glivenko-Cantelli*. Hence

$$\sup_{\beta \in \mathbb{R}_A^p \times \mathcal{M}_A} |L_n(\beta) - L_0(\beta)| \xrightarrow{p} 0, \text{ as } n \rightarrow \infty, \quad (1)$$

where $L_0(\beta) = \mathbb{E}\rho_\tau(Y - X^\top \theta - m(Z))$.

By Assumption 1 – 3, a similar proof for Lemma 3.3 in He and Shi (1994) implies that, for any $\epsilon > 0$,

$$\inf_{\substack{d(\beta, \beta_0) > \epsilon, \\ \beta \in \mathbb{R}_A^p \times \mathcal{M}_A}} L_0(\beta) > L_0(\beta_0). \quad (2)$$

For the true function m_0 , let

$$m^* = \arg \min_{m \in \mathcal{M}_A} \|m - m_0\|_{L^2([0,1]^q)} \text{ and } \beta^* = (\theta_0, m^*).$$

Then, we have $L_n(\hat{\beta}) \leq L_n(\beta^*)$ by the definition of $\hat{\beta}$. This and (1), (2) imply that

$$d(\hat{\beta}, \beta^*) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

On the other hand, by equation (26) in Schmidt-Hieber (2020), we have

$$\|m^* - m_0\|_{L^2([0,1]^q)} = O(r_n). \quad (3)$$

It follows that

$$d(\hat{\beta}, \beta_0) \leq d(\hat{\beta}, \beta^*) + \|m^* - m_0\|_{L^2([0,1]^q)} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

This completes the proof of the consistency of $\hat{\beta}$.

Next we prove $d(\hat{\beta}, \beta_0) = O_p(r_n \log^2 n)$. Write $R = L \prod_{k=0}^L (q_k + 1) \sum_{k=1}^L q_{k-1} q_k$ and

$$\mathcal{A}_\delta = \{\beta \in \mathbb{R}_A^p \times \mathcal{M}(s, L, \mathbf{q}, A) \mid \delta/2 \leq d(\beta, \beta^*) \leq \delta\}. \quad (4)$$

We verify that, for any $\delta > 0$,

$$\mathbb{E}^*[\sup_{\beta \in \mathcal{A}_\delta} \sqrt{n}\{(L_n - L_0)(\beta^*) - (L_n - L_0)(\beta)\}] \lesssim \phi_n(\delta), \quad (5)$$

where $\mathbb{E}^*$ is an outer measure, $\phi_n(\delta) = \delta \sqrt{s \log \frac{R}{\delta}} + \frac{s}{\sqrt{n}} \log \frac{R}{\delta}$, and $a_n \lesssim b_n$ means $a_n \leq cb_n$ for some constant $c > 0$.

Denote $\rho_\tau(\beta) = \rho_\tau(Y - X^\top \theta - m(Z))$ and $\mathcal{B}_\delta = \{\rho_\tau(\beta^*) - \rho_\tau(\beta) \mid \beta \in \mathcal{A}_\delta\}$. For any $\beta, \beta_1 \in \mathcal{A}_\delta$, we have $\mathbb{E}|\rho_\tau(\beta) - \rho_\tau(\beta_1)|^2 \leq 4d^2(\beta, \beta_1)$. Lemma 5 in Schmidt-Hieber (2020) then implies that

$$\log(1 + \mathcal{N}_{[\cdot]}(\epsilon, \mathcal{B}_\delta, L^2(P))) \lesssim s \log \frac{R}{\epsilon},$$

where $\mathcal{N}_{[\cdot]}(\epsilon, \mathcal{B}_\delta, L^2(P))$ is the bracket number of $\mathcal{B}_\delta$ with $L^2(P)$ norm. It follows that

$$J_{[\cdot]}(\delta, \mathcal{B}_\delta) = \int_0^\delta \sqrt{1 + \mathcal{N}_{[\cdot]}(\epsilon, \mathcal{B}_\delta, L^2(P))} d\epsilon \lesssim \delta \sqrt{s \log \frac{R}{\delta}}.$$

By Lemma 3.4.2 in Van der Vaart and Wellner (1996), we conclude that

$$\begin{aligned} \mathbb{E}^*[\sup_{\beta \in \mathcal{A}_\delta} \sqrt{n}\{(L_n - L_0)(\beta^*) - (L_n - L_0)(\beta)\}] &= \mathbb{E}^*[\sup_{\beta \in \mathcal{A}_\delta} \sqrt{n}(\mathbb{P}_n - \mathbb{P})\{\rho_\tau(\beta^*) - \rho_\tau(\beta)\}] \\ &\lesssim J_{[\cdot]}(\delta, \mathcal{B}_\delta) \left\{ \frac{J_{[\cdot]}(\delta, \mathcal{B}_\delta)}{\delta^2 \sqrt{n}} + 1 \right\} \\ &= \phi_n(\delta). \end{aligned}$$

Let $\eta_n = r_n \log^2 n$. It is clear that

$$\frac{1}{\eta_n^2} \phi_n(\eta_n) \lesssim \sqrt{n} \text{ and } L_n(\hat{\beta}) \leq L_n(\beta^*). \quad (6)$$

Then with (5), (6) and Theorem 3.4.1 in Van der Vaart and Wellner (1996), we have

$$d(\hat{\beta}, \beta^*) = O_p(\eta_n).$$

Moreover, by Assumption 3 and the definition of φ^* , we have

$$\begin{aligned} d^2(\hat{\beta}, \beta_0) &= \mathbb{E}\{X^\top(\hat{\theta} - \theta_0) + \hat{m}(Z) - m_0(Z)\}^2 \\ &\geq \frac{1}{c_0} \mathbb{E}[f(0|U)\{X^\top(\hat{\theta} - \theta_0) + \hat{m}(Z) - m_0(Z)\}^2] \\ &= \frac{1}{c_0} \mathbb{E}[f(0|U)\{(X - \varphi^*(Z))^\top(\hat{\theta} - \theta_0) + (\hat{\theta} - \theta_0)^\top \varphi^*(Z) + \hat{m}(Z) - m_0(Z)\}^2] \\ &= \frac{1}{c_0} \mathbb{E}[f(0|U)\{(X - \varphi^*(Z))^\top(\hat{\theta} - \theta_0)\}^2] \\ &\quad + \frac{1}{c_0} \mathbb{E}[f(0|U)\{(\hat{\theta} - \theta_0)^\top \varphi^*(Z) + \hat{m}(Z) - m_0(Z)\}^2]. \end{aligned}$$

Since the matrix $\mathbb{E}[f(0|U)\{X - \varphi^*(Z)\}\{X - \varphi^*(Z)\}^\top]$ is positive definite, it follows that

$\|\hat{\theta} - \theta_0\| = O_p(r_n \log^2 n)$ and thus $\|\hat{m} - m_0\|_{L^2([0,1]^q)} = O_p(r_n \log^2 n)$. This completes the proof.

Proof of Theorem 3.2. For simplicity, we only consider the proof for the median quantile regression case, when $\tau = 0.5$. To derive the minimax lower bound, it suffices to show that,

when the error ϵ is the standard normal distribution and the parameter θ_0 is known and fixed, there exists a subset $\mathcal{H}^*$ of $\mathcal{H}(J, \gamma, \mathbf{d}, \bar{\mathbf{d}}, B)$ in Assumption 1, such that

$$\lim_{C \rightarrow \infty} \liminf_{n \rightarrow \infty} \sup_{\hat{m}} \sup_{m_0 \in \mathcal{H}^*} \mathbb{P}_{(\theta_0, m_0, f)}(\|\hat{m} - m_0\|_{L^2([0,1]^q)} \geq Cr_n) = 1, \quad (7)$$

where f is the probability density function of a standard normal distribution. Then this reduces to a nonparametric regression with normal errors. Hence, (7) holds by Theorem 3 in Schmidt-Hieber (2020).

Proof of Theorem 3.3. For $\hat{\beta} = (\hat{\theta}, \hat{m})$, we write $\zeta = \theta - \theta_0$, $\hat{\zeta} = \hat{\theta} - \theta_0$, $h(Z) = m(Z) - m_0(Z) + (\theta - \theta_0)^\top \varphi^*(Z)$, $\hat{h}(Z) = \hat{m}(Z) - m_0(Z) + (\hat{\theta} - \theta_0)^\top \varphi^*(Z)$ and $\tilde{X} = X - \varphi^*(Z)$. These imply that

$$\frac{1}{n} \sum_{i=1}^n \rho_\tau(Y_i - \theta^\top X - m(Z)) = \frac{1}{n} \sum_{i=1}^n \rho_\tau(\epsilon_i - \zeta^\top \tilde{X}_i - h(Z_i)).$$

Denote $L_n(\zeta, h) = \frac{1}{n} \sum_{i=1}^n \rho_\tau(\epsilon_i - \zeta^\top \tilde{X}_i - h(Z_i))$. We define the subgradient of the loss function L_n at ζ as

$$\Psi_n(\zeta, h) = \frac{\partial L_n(\zeta, h)}{\partial \zeta} = \mathbb{P}_n \psi_\tau(\tilde{X}, Z; \zeta, h),$$

where $\psi_\tau(\tilde{X}, Z; \zeta, h) = -\{\tau - 1(\epsilon - \zeta^\top \tilde{X} - h(Z) < 0)\} \tilde{X}$. Let $\Psi_0(\zeta, h) = \mathbb{E} \psi_\tau(\tilde{X}, Z; \zeta, h)$.

With $\mathcal{A}_\delta$ defined in (4), we further define $\tilde{\mathcal{A}}_\delta = \{(\zeta, h) \mid \zeta = \theta - \theta_0, h(Z) = m(Z) - m_0(Z) + (\theta - \theta_0)^\top \varphi^*(Z), (\theta, m) \in \mathcal{A}_\delta\}$ and $\mathcal{C}_\delta = \{\psi_\tau(\zeta, h) - \psi_\tau(\zeta_0, h_0) \mid (\zeta, h) \in \tilde{\mathcal{A}}_\delta \text{ and } (\zeta_0, h_0) = (0, 0)\}$. Then in analogy to the proof of Theorem 3.1, we have, for any $\delta > 0$,

$$J_{[\cdot]}(\delta, \mathcal{C}_\delta) = \int_0^\delta \sqrt{1 + \mathcal{N}_{[\cdot]}(\epsilon, \mathcal{C}_\delta, L^2(P))} d\epsilon \lesssim \delta \sqrt{s \log \frac{R}{\delta}}.$$

Thus it follows Lemma 3.4.2 in Van der Vaart and Wellner (1996), then,

$$\begin{aligned}
& \mathbb{E}^* \left\{ \sup_{(\zeta, h) \in \mathcal{C}_\delta} \left| \sqrt{n} [(\Psi_n - \Psi_0)(\zeta, h) - (\Psi_n - \Psi_0)(\zeta_0, h_0)] \right| \right\} \\
&= \mathbb{E}^* \left\{ \sup_{(\zeta, h) \in \mathcal{C}_\delta} \left| \sqrt{n} (\mathbb{P}_n - \mathbb{P}) [\psi_\tau(\tilde{X}, Z; \zeta, h) - \psi_\tau(\tilde{X}, Z; \zeta_0, h_0)] \right| \right\} \\
&\lesssim J_{[\cdot]}(\delta, \mathcal{C}_\delta) \left\{ \frac{J_{[\cdot]}(\delta, \mathcal{C}_\delta)}{\delta^2 \sqrt{n}} + 1 \right\} \\
&= \phi_n(\delta).
\end{aligned}$$

Since $(\hat{\zeta}, \hat{h})$ converges to (ζ_0, h_0) at rate $r_n \log^2 n$, then with $\delta_n = r_n \log^2 n$,

$$\left| \sqrt{n} [(\Psi_n - \Psi_0)(\zeta, h)|_{(\zeta, h) = (\hat{\zeta}, \hat{h})} - (\Psi_n - \Psi_0)(\zeta_0, h_0)] \right| \lesssim \phi_n(\delta_n) = o_p(1),$$

or, written alternatively,

$$\sqrt{n} \{ \Psi_0(\zeta, h)|_{(\zeta, h) = (\hat{\zeta}, \hat{h})} + \Psi_n(\zeta_0, h_0) \} = \sqrt{n} \{ \Psi_n(\zeta, h)|_{(\zeta, h) = (\hat{\zeta}, \hat{h})} + \Psi_0(\zeta_0, h_0) \} + o_p(1). \quad (8)$$

Let $\tilde{Y}_i = \epsilon_i - \hat{h}(Z_i)$, $i = 1, \dots, n$. Then $\hat{\zeta}$ is the minimizer of $L_n^*(\zeta) = \frac{1}{n} \sum_{i=1}^n \rho_\tau(\tilde{Y}_i - \zeta^\top \tilde{X}_i)$ with respect to ζ and

$$\Psi_n(\zeta, h)|_{(\zeta, h) = (\hat{\zeta}, \hat{h})} = \frac{dL_n^*(\zeta)}{d\zeta} \Big|_{\zeta = \hat{\zeta}} = \frac{1}{n} \sum_{i=1}^n -\{\tau - 1(\tilde{Y}_i - \hat{\zeta}^\top \tilde{X}_i < 0)\} \tilde{X}_i$$

Since L_n^* is a continuous piecewise function of ζ , it follows that the subgradient is bounded by the difference between the right and left derivatives. Thus,

$$\begin{aligned}
\left| \frac{dL_n^*(\zeta)}{d\zeta} \Big|_{\zeta = \hat{\zeta}} \right| &\leq \frac{2}{n} \sum_{i=1}^n 1(\tilde{Y}_i = \hat{\zeta}^\top \tilde{X}_i) |\tilde{X}_i| \\
&\leq \left\{ 2 \sum_{i=1}^n 1(\tilde{Y}_i = \hat{\zeta}^\top \tilde{X}_i) \right\} \max_{i=1, \dots, n} \left(\frac{|\tilde{X}_i|}{n} \right) \\
&= o_p\left(\frac{1}{\sqrt{n}}\right),
\end{aligned}$$

where $|\cdot|$ and $\max(\cdot)$ operate component-wise on vector and the last equality holds due to Assumption 2, Assumption 6 and the fact $\sum_{i=1}^n 1(\tilde{Y}_i = \hat{\zeta}^\top \tilde{X}_i) \leq p$ by Lemma 2.1 in El-Attar et al. (1979) and Sard's theorem in Milnor (1965). Moreover, a calculation yields $\Psi_0(\zeta_0, h_0) = 0$, so the left hand side of (8) satisfies

$$\sqrt{n} \{ \Psi_0(\zeta, h)|_{(\zeta, h) = (\hat{\zeta}, \hat{h})} + \Psi_n(\zeta_0, h_0) \} = o_p(1),$$

or equivalently,

$$\sqrt{n}\Psi_0(\zeta, h)|_{(\zeta, h)=(\hat{\zeta}, \hat{h})} = -\sqrt{n}\Psi_n(\zeta_0, h_0) + o_p(1).$$

On the other hand, applying the Taylor's expansion for $\Psi_0(\zeta, h)|_{(\zeta, h)=(\hat{\zeta}, \hat{h})}$ at (ζ_0, h_0) , we obtain

$$\Psi_0(\zeta, h)|_{(\zeta, h)=(\hat{\zeta}, \hat{h})} = 2\mathbb{E}\{f_\tau(0|U)\tilde{X}\tilde{X}^\top\}(\hat{\zeta} - \zeta_0) + O(d^2(\hat{\beta}, \beta_0)).$$

Here the derivative with respect to h is based on the derivative of some smooth curve $\{h_{(t)} : t \in \mathbb{R}, h_{(0)} = h_0 \text{ and } h_{(1)} = \hat{h}\}$ with respect to t . Since $\hat{\zeta} - \zeta_0 = \hat{\theta} - \theta_0$ and $\bar{\gamma}_{\bar{k}} > \bar{d}_{\bar{k}}/2$. It follows that

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{2}[\mathbb{E}\{f_\tau(0|U)\tilde{X}\tilde{X}^\top\}]^{-1}\sqrt{n}\Psi_n(\zeta_0, h_0) + o_p(1) \rightarrow N(0, \Sigma_2^{-1}\Sigma_1\Sigma_2^{-1}).$$

Therefore, the result follows.

B Additional Numerical Results

B.1 Simulations

Table 5 - Table 7 show the additional simulation results with homoscedastic random errors in Section 5.1. Table 8 - Table 11 show the numerical results for the simulation with heteroscedastic random errors in Section 5.2. Table 12 is the numerical results under different proportion of the neural weights being zero.

Table 5: Bias and standard deviation (in parentheses) of $\hat{\theta}_1$ for the LQR, PLAQR and DPLQR methods under homoscedastic random errors.

θ	Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
			LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
θ_2	Case 1 (linear)	1000	0.0785 (0.3106)	0.0820 (0.3246)	0.1194 (0.3373)	-0.0119 (0.2826)	0.0307 (0.3038)	0.0643 (0.3192)	-0.0658 (0.2977)	-0.0729 (0.3278)	0.1218 (0.3315)
			0.0638 (0.2246)	0.0731 (0.2298)	0.0886 (0.2317)	-0.0104 (0.1949)	0.0274 (0.2184)	0.0419 (0.2236)	-0.0543 (0.2394)	0.0607 (0.2429)	0.0653 (0.2558)
		2000	0.0763 (0.3547)	0.0281 (0.2494)	0.0453 (0.2605)	0.0593 (0.2708)	-0.0183 (0.1943)	0.0313 (0.2005)	-0.0884 (0.3425)	-0.0277 (0.2448)	0.0482 (0.2522)
			0.0697 (0.2280)	-0.0135 (0.1746)	0.0334 (0.1899)	-0.0527 (0.1984)	-0.0212 (0.1329)	0.0262 (0.1506)	-0.0735 (0.2386)	-0.0104 (0.1636)	0.0319 (0.1781)
	Case 2 (additive)	1000	0.1188 (0.3285)	-0.0929 (0.3086)	0.0672 (0.2702)	-0.1009 (0.2699)	-0.0884 (0.2503)	0.0445 (0.2116)	-0.1256 (0.3291)	-0.0998 (0.3108)	0.0615 (0.2882)
			0.1052 (0.2231)	0.0878 (0.2106)	0.0556 (0.1986)	-0.0933 (0.1869)	-0.0675 (0.1773)	0.0381 (0.1496)	-0.1183 (0.2297)	0.0893 (0.2212)	0.0405 (0.2015)

Table 6: Empirical coverage probability of the 95% confidence interval for θ_2 by the LQR, PLAQR and DPLQR methods under homoscedastic random errors.

θ	Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
			LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
θ_2	Case 1	1000	0.910	0.890	0.885	0.935	0.910	0.900	0.925	0.895	0.895
	(linear)	2000	0.935	0.910	0.915	0.940	0.920	0.925	0.930	0.925	0.925
	Case 2	1000	0.890	0.910	0.895	0.910	0.925	0.915	0.915	0.920	0.910
	(additive)	2000	0.910	0.935	0.920	0.935	0.945	0.940	0.915	0.930	0.925
	Case 3	1000	0.895	0.900	0.910	0.910	0.920	0.925	0.905	0.905	0.915
	(deep)	2000	0.910	0.920	0.930	0.925	0.930	0.940	0.915	0.915	0.935

Table 7: Empirical coverage probability of the 95% confidence interval for $\theta = (\theta_1, \theta_2)$ and mean of the variance estimates (in parentheses) of $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2)$ for the proposed and bootstrap methods for Case 3 (deep) under homoscedastic random errors. The results are based on 200 simulation runs with 300 bootstraps per run. PM: proposed method, BM: bootstrap method.

θ	n	$\tau = 0.25$		$\tau = 0.50$		$\tau = 0.75$	
		PM	BM	PM	BM	PM	BM
θ_1	1000	0.910	0.865	0.930	0.895	0.925	0.875
		(0.1290)	(0.1189)	(0.1143)	(0.0982)	(0.1293)	(0.1168)
	2000	0.935	0.905	0.955	0.910	0.945	0.900
		(0.0909)	(0.0862)	(0.0797)	(0.0716)	(0.0921)	(0.0816)
θ_2	1000	0.910	0.825	0.925	0.850	0.915	0.805
		(0.2477)	(0.2321)	(0.2193)	(0.2063)	(0.2480)	(0.2373)
	2000	0.930	0.850	0.940	0.870	0.935	0.855
		(0.1751)	(0.1614)	(0.1535)	(0.1439)	(0.1772)	(0.1676)

Table 8: Bias and standard deviation (in parentheses) of $\hat{\theta}$ for the LQR, PLAQR and DPLQR methods under heteroscedastic random errors.

θ	Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
			LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
θ_1	Case 4 (linear)	1000	0.0088 (0.0688)	-0.0097 (0.0721)	0.0122 (0.0762)	-0.0082 (0.0543)	-0.0095 (0.0633)	0.0110 (0.0647)	-0.0094 (0.0617)	-0.0127 (0.0701)	0.0132 (0.0735)
			-0.0062 (0.0453)	0.0074 (0.0583)	0.0087 (0.0613)	-0.0050 (0.0367)	0.0076 (0.0486)	0.0086 (0.0502)	-0.0054 (0.0446)	-0.0070 (0.0562)	0.0085 (0.0605)
		2000	0.0219 (0.0573)	0.0170 (0.0427)	0.0183 (0.0523)	-0.0199 (0.0422)	-0.0154 (0.0345)	-0.0176 (0.0401)	-0.0205 (0.0543)	-0.0178 (0.0396)	-0.0181 (0.0495)
			-0.0181 (0.0408)	0.0159 (0.0317)	0.0169 (0.0353)	-0.0152 (0.0265)	-0.0132 (0.0255)	-0.0141 (0.0261)	-0.0172 (0.0388)	-0.0149 (0.0298)	-0.0155 (0.0339)
	Case 5 (additive)	1000	-0.0109 (0.0812)	0.0097 (0.0733)	0.0071 (0.0585)	-0.0107 (0.0671)	-0.0100 (0.0631)	0.0036 (0.0453)	-0.0131 (0.0820)	-0.0124 (0.0776)	0.0035 (0.0554)
			-0.0087 (0.0610)	0.0076 (0.0584)	0.0031 (0.0428)	-0.0101 (0.0505)	-0.0092 (0.0456)	0.0028 (0.0334)	-0.0113 (0.0597)	-0.0105 (0.0512)	0.0036 (0.0424)
	Case 6 (deep)	1000	0.0091 (0.1208)	0.0103 (0.1239)	0.0134 (0.1342)	0.0072 (0.0918)	-0.0094 (0.0971)	0.0116 (0.1098)	-0.0095 (0.1096)	-0.0110 (0.1144)	0.0142 (0.1319)
			0.0059 (0.0818)	-0.0079 (0.0876)	0.0101 (0.0927)	0.0047 (0.0710)	-0.0061 (0.0735)	0.0096 (0.0802)	0.0064 (0.0791)	0.0075 (0.0841)	0.0113 (0.0905)
		2000	-0.0416 (0.1097)	0.0173 (0.0707)	0.0258 (0.0953)	-0.0385 (0.0871)	-0.0128 (0.0614)	0.0209 (0.0754)	-0.0454 (0.0914)	-0.0208 (0.0684)	0.0388 (0.0821)
			0.0339 (0.0756)	0.0137 (0.0517)	0.0186 (0.0613)	-0.0290 (0.0578)	-0.0093 (0.0411)	0.0128 (0.0484)	-0.0358 (0.0614)	-0.0142 (0.0467)	0.0216 (0.0635)
	Case 5 (additive)	1000	0.0684 (0.1410)	0.0585 (0.1093)	0.0454 (0.0990)	-0.0536 (0.1124)	-0.0437 (0.0898)	0.0355 (0.0815)	0.0656 (0.1416)	0.0582 (0.1244)	0.0484 (0.1012)
			0.0578 (0.0886)	0.0503 (0.0874)	0.0388 (0.0609)	-0.0413 (0.0826)	-0.0309 (0.0732)	0.0247 (0.0596)	-0.0538 (0.0928)	-0.0464 (0.0894)	0.0364 (0.0701)
θ_2	Case 4 (linear)	1000	0.0091 (0.1208)	0.0103 (0.1239)	0.0134 (0.1342)	0.0072 (0.0918)	-0.0094 (0.0971)	0.0116 (0.1098)	-0.0095 (0.1096)	-0.0110 (0.1144)	0.0142 (0.1319)
			0.0059 (0.0818)	-0.0079 (0.0876)	0.0101 (0.0927)	0.0047 (0.0710)	-0.0061 (0.0735)	0.0096 (0.0802)	0.0064 (0.0791)	0.0075 (0.0841)	0.0113 (0.0905)
		2000	-0.0416 (0.1097)	0.0173 (0.0707)	0.0258 (0.0953)	-0.0385 (0.0871)	-0.0128 (0.0614)	0.0209 (0.0754)	-0.0454 (0.0914)	-0.0208 (0.0684)	0.0388 (0.0821)
			0.0339 (0.0756)	0.0137 (0.0517)	0.0186 (0.0613)	-0.0290 (0.0578)	-0.0093 (0.0411)	0.0128 (0.0484)	-0.0358 (0.0614)	-0.0142 (0.0467)	0.0216 (0.0635)
	Case 5 (additive)	1000	0.0684 (0.1410)	0.0585 (0.1093)	0.0454 (0.0990)	-0.0536 (0.1124)	-0.0437 (0.0898)	0.0355 (0.0815)	0.0656 (0.1416)	0.0582 (0.1244)	0.0484 (0.1012)
			0.0578 (0.0886)	0.0503 (0.0874)	0.0388 (0.0609)	-0.0413 (0.0826)	-0.0309 (0.0732)	0.0247 (0.0596)	-0.0538 (0.0928)	-0.0464 (0.0894)	0.0364 (0.0701)

Table 9: Empirical coverage probability of the 95% confidence interval for $\theta = (\theta_1, \theta_2)$ by the LQR, PLAQR and DPLQR methods under heteroscedastic random errors.

θ	Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
			LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
θ_1	Case 4	1000	0.905	0.900	0.875	0.920	0.910	0.900	0.900	0.900	0.885
	(linear)	2000	0.925	0.915	0.915	0.935	0.920	0.925	0.930	0.925	0.915
	Case 5	1000	0.890	0.910	0.910	0.905	0.920	0.915	0.900	0.915	0.910
	(additive)	2000	0.915	0.935	0.930	0.920	0.945	0.935	0.910	0.950	0.925
	Case 6	1000	0.890	0.905	0.915	0.905	0.905	0.920	0.900	0.910	0.920
	(deep)	2000	0.910	0.915	0.935	0.910	0.925	0.940	0.910	0.920	0.960
θ_2	Case 4	1000	0.900	0.895	0.885	0.915	0.895	0.890	0.915	0.910	0.900
	(linear)	2000	0.925	0.910	0.915	0.935	0.925	0.930	0.930	0.915	0.910
	Case 5	1000	0.900	0.915	0.905	0.915	0.925	0.915	0.915	0.920	0.910
	(additive)	2000	0.915	0.925	0.920	0.920	0.935	0.925	0.920	0.935	0.925
	Case 6	1000	0.895	0.905	0.920	0.910	0.910	0.925	0.900	0.910	0.920
	(deep)	2000	0.910	0.920	0.935	0.915	0.920	0.940	0.915	0.925	0.940

Table 10: Empirical coverage probability of the 95% confidence interval for $\theta = (\theta_1, \theta_2)$ and mean of the variance estimates (in parentheses) of $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2)$ for the proposed and bootstrap methods for Case 6 (deep) under heteroscedastic random errors. The results are based on 200 simulation runs with 300 bootstraps per run. PM: proposed method, BM: bootstrap method.

θ	n	$\tau = 0.25$		$\tau = 0.50$		$\tau = 0.75$	
		PM	BM	PM	BM	PM	BM
θ_1	1000	0.915	0.910	0.920	0.925	0.920	0.910
		(0.0515)	(0.0538)	(0.0473)	(0.0453)	(0.0522)	(0.0515)
	2000	0.935	0.965	0.940	0.965	0.960	0.930
		(0.0416)	(0.0403)	(0.0366)	(0.0343)	(0.0425)	(0.0413)
θ_2	1000	0.920	0.865	0.925	0.885	0.920	0.875
		(0.0989)	(0.0849)	(0.0908)	(0.0774)	(0.1001)	(0.0890)
	2000	0.935	0.900	0.940	0.920	0.940	0.910
		(0.0654)	(0.0647)	(0.0607)	(0.0593)	(0.0683)	(0.0654)

Table 10: Relative mean squared error of $\hat{m}$ for the LQR, PLAQR and DPLQR methods under heteroscedastic random errors.

Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
		LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
Case 4	1000	0.0024	0.0032	0.0048	0.0011	0.0021	0.0028	0.0023	0.0032	0.0038
(linear)	2000	0.0011	0.0021	0.0030	0.0005	0.0009	0.0015	0.0011	0.0019	0.0026
Case 5	1000	0.0039	0.0035	0.0038	0.0030	0.0025	0.0026	0.0035	0.0029	0.0031
(additive)	2000	0.0030	0.0023	0.0027	0.0024	0.0020	0.0022	0.0029	0.0026	0.0028
Case 6	1000	0.0224	0.0205	0.0142	0.0154	0.0147	0.0093	0.0208	0.0179	0.0123
(deep)	2000	0.0198	0.0196	0.0105	0.0144	0.0144	0.0062	0.0153	0.0143	0.0098

Table 11: Mean and standard deviation (in parentheses) of the excess risk to evaluate prediction for the LQR, PLAQR and DPLQR methods under heteroscedastic random errors.

Case	n	$\tau = 0.25$			$\tau = 0.50$			$\tau = 0.75$		
		LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR	LQR	PLAQR	DPLQR
Case 4 (linear)	1000	0.0021	0.0028	0.0032	0.0025	0.0030	0.0035	0.0021	0.0029	0.0033
		(0.0011)	(0.0022)	(0.0024)	(0.0015)	(0.0026)	(0.0030)	(0.0012)	(0.0022)	(0.0023)
	2000	0.0010	0.0014	0.0021	0.0012	0.0019	0.0025	0.0011	0.0014	0.0019
		(0.0006)	(0.0014)	(0.0016)	(0.0007)	(0.0016)	(0.0020)	(0.0007)	(0.0015)	(0.0016)
Case 5 (additive)	1000	0.0169	0.0093	0.0133	0.0189	0.0102	0.0142	0.0158	0.0097	0.0125
		(0.0034)	(0.0023)	(0.0029)	(0.0037)	(0.0026)	(0.0032)	(0.0036)	(0.0022)	(0.0029)
	2000	0.0160	0.0083	0.0098	0.0168	0.0091	0.0108	0.0139	0.0071	0.0090
		(0.0028)	(0.0019)	(0.0021)	(0.0030)	(0.0021)	(0.0023)	(0.0029)	(0.0021)	(0.0022)
Case 6 (deep)	1000	0.0147	0.0122	0.0111	0.0201	0.0154	0.0139	0.0182	0.0115	0.0112
		(0.0045)	(0.0037)	(0.0033)	(0.0048)	(0.0039)	(0.0034)	(0.0051)	(0.0048)	(0.0032)
	2000	0.0139	0.0109	0.0086	0.0195	0.0135	0.0106	0.0175	0.0102	0.0088
		(0.0039)	(0.0031)	(0.0023)	(0.0042)	(0.0033)	(0.0026)	(0.0038)	(0.0033)	(0.0023)

Table 12: Relative mean square error of $\hat{m}$ under Case 3 (deep) and Case 6 (deep) with ω (percentage) of weights being zero.

n	Case 3 (deep)			Case 6 (deep)		
	$\omega=30\%$	$\omega=50\%$	$\omega=70\%$	$\omega=30\%$	$\omega=50\%$	$\omega=70\%$
1000	0.0208	0.0209	0.0218	0.0091	0.0093	0.0097
2000	0.0155	0.0159	0.0161	0.0060	0.0062	0.0069

B.2 Additional Data Applications

We also apply the proposed method to two more data set, i.e., Boston Housing and Concrete Compressive Strength data.

B.2.1 Concrete Compressive Strength Data

We apply the proposed methodology, along with the competing methods, to the Concrete Compressive Strength Data Set (Yeh, 1998) available on the UCI machine learning repository. The data consist of $n = 1030$ observations, with the response being a continuous variable of concrete compressive strength (CCS), and eight covariates: cement (Z_1), water (Z_2), fly ash (Z_3), blast furnace slag (Z_4), superplasticizer (Z_5), coarse aggregate (Z_6), fine aggregate (Z_7) and age of the mixture in days (Z_8), of which the first seventh covariates are the ingredients in high-performance concrete (HPC). For concrete technology, the water-cement ratios (WCR) has been recognized as the most useful and significant advancement for CCS (Yeh, 1998). Here we not only explore the association between WCR and CCS, but also predict the CCS of HPC from the covariates.

Similar to the analysis in Natality Birth and Boston Housing Data, we model the data with four approaches: LQR, PLAQR, DPLQR and DNQR. Here we treat log CCS as the

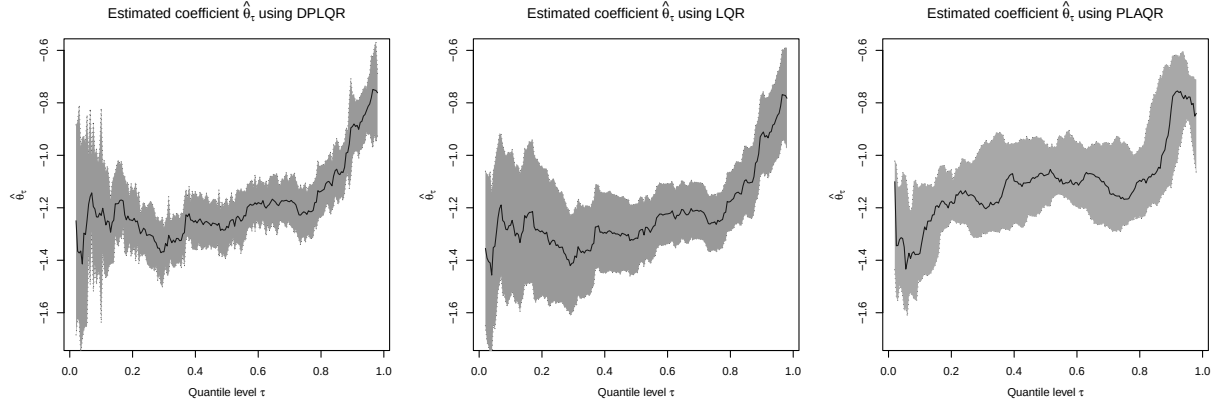


Figure 5: Estimation for θ_τ with $\tau \in [0.02, 0.98]$ and 95% confidence interval using DPLQR (left panel), LQR (middle panel) and PLAQR (right panel) in Concrete Compressive Strength Data.

response, WCR, i.e, Z_2/Z_1 , as the linear predictor X . Figure 5 with the parametric estimates and the 95% confidence interval under LQR, PLAQR, DPLQR, suggests that there is a strong association between WCR and CCS. The negative estimates further support the Abrams rule in civil engineering that increase in the WCR tends to decrease the strength of concrete (Gorse et al., 2012). Among the first three approaches, DPLQR produced narrower 95% confidence intervals area (0.2891) for the parameter a than the LQR (0.3408) and PLAQR (0.3137). The prediction results in Figure 6 further reveal that our method not only improves the prediction accuracy compared to LQR and PLAQR. In addition, although the proposed model is a submodel of the DNQR, its performance in prediction is comparable. Thus, compared to the fully nonparametric approach (DNQR), the partially linear approach (DPLQR) trades a small amount of prediction accuracy for interpretability and has the best performance among the three interpretable approaches, i.e., LQR, PLAQR and DPLQR.

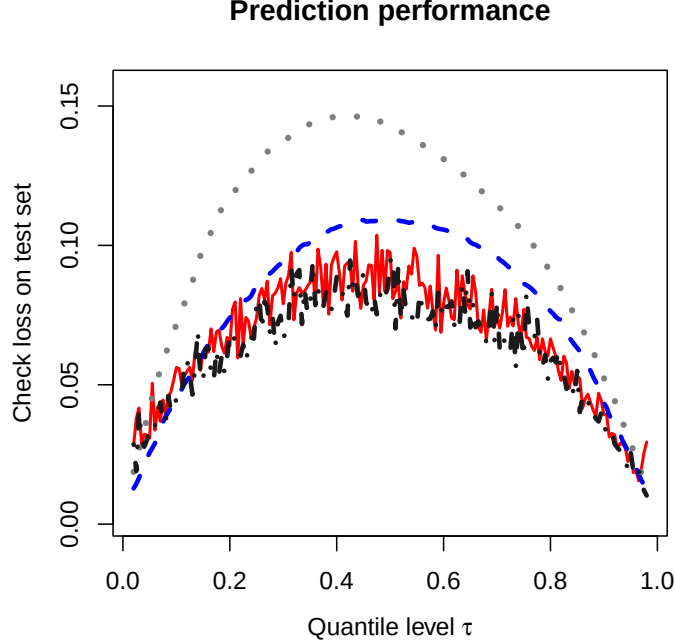


Figure 6: Check loss evaluated on test set with quantile levels $\tau \in [0.02, 0.98]$ under four models (LQR, PLAQR, DPLQR and DNQR) in Concrete Compressive Strength Data. Dotted line: LQR (ACL=0.1052); dashed line: PLAQR (ACL=0.0786); solid line: DPLQR (ACL=0.0675); dot-dashed line: DNQR (ACL=0.0626).

B.2.2 Boston Housing Data

The Boston Housing Data, available from the R package *mlbench*, is a benchmark dataset for quantile regression analysis. In Harrison and Rubinfeld (1978), 506 observations were examined to study the housing prices based on various demographic and socioeconomic predictors. The variables are: median house price (Y), per capita crime rate by town (X_1), a river boundary indicator (X_2), proportion of non-retail business acres per town (X_3), proportion of residential land zoned for lots (X_4), nitrogen oxides concentration (X_5), average number of rooms per dwelling (X_6), proportion of owner-occupied units built prior to 1940 (X_7), weighted mean of distances to five Boston employment centers (X_8), index

of accessibility to radial highways (X_9), full-value property-tax rate (X_{10}), pupil-teacher ratio by town (X_{11}), the proportion of black individuals by town (X_{12}), the percentage of the population classified as lower status (X_{13}).

Since the work of Thaler (1978), economists have tried to understand the impact of crime rate on housing price. To address this question, we model X_1 (per capita crime rate by town) as the parametric predictor to study the effect of the crime rates on house price. All other covariates are treated as the nonparametric predictors, i.e.,

$$\log Y = \theta_1 X_1 + m(X_2, X_3, \dots, X_{13}) + \epsilon.$$

We also consider the linear quantile regression (LQR), partially linear additive quantile regression (PLAQR) and the deep nonparametric quantile regression [DNQR, see Padilla et al. (2020) and Shen et al. (2021)], which treats all thirteen covariates as components of a nonparametric regression model, i.e., $\log Y = m(X_1, X_2, X_3, \dots, X_{13}) + \epsilon$. Note that DNQR does not offer direct treatment effects.

The estimates $\hat{\theta}_1$ from the quantile levels $\tau \in [0.02, 0.98]$ are shown in Figure 7, revealing that the crime rate has a significant effect on the price of a house and house prices are higher in areas with lower crime rates. Figure 8 displays the prediction error in terms of check loss on randomly hold-out test set (20% of data). It shows the proposed method is considerably better than the LQR and PLADR methods and is comparable to DNQR.

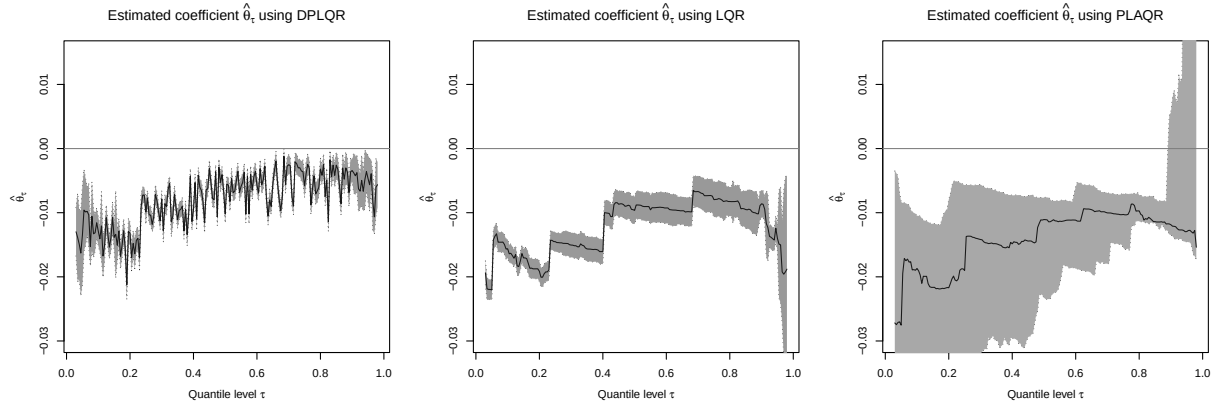


Figure 7: Estimation for θ_τ with $\tau \in [0.02, 0.98]$ and 95% confidence interval using DPLQR (left panel), LQR (middle panel) and PLAQR (right panel) in Boston Housing Data.

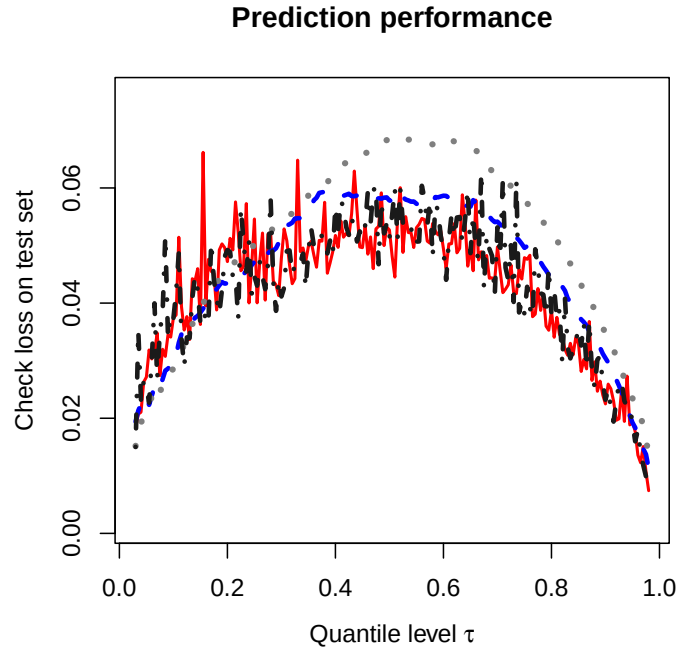


Figure 8: Check loss evaluated on test set with quantile levels $\tau \in [0.02, 0.98]$ under four models (LQR, PLAQR, DPLQR and DNQR) in Boston Housing Data. Dotted line: LQR (ACL=0.0508); dashed line: PLAQR LQR (ACL=0.0458); solid line: DPLQR (ACL=0.0425); dot-dashed line: DNQR (ACL=0.0430). Solid line and dot-dashed line almost overlap.

C Tuning Parameters

Based on the check loss on validation set, we used grid search to choose tuning parameters among a set of candidates. The candidates for simulations and data analysis are given in Table 13 and 14, respectively.

Table 13: Candidates of tuning parameters in the simulations

Tunning parameters	Possible values
Depth	$\{1, 2, 3, 5\}$
Width	$\{16, 32, 64\}$
Epoch	$\{200, 500\}$
Minibatch	$\{64, 128\}$
Early stop	$\{200, 300\}$
Learning rate	$\{0.001, 0.005, 0.01, 0.05\}$

Table 14: Candidates of tuning parameters in the data analysis

Tunning parameters	Possible values
Depth	$\{2, 3, 5, 8, 10\}$
Width	$\{64, 128, 256\}$
Epoch	$\{500, 600\}$
Minibatch	$\{128, 256\}$
Early stop	$\{200, 300, 500\}$
Learning rate	$\{0.0001, 0.0005, 0.001, 0.01\}$

References

- El-Attar, R. A., Vidyasagar, M., and Dutta, S. R. K. (1979), “An algorithm for l_1 -norm minimization with application to nonlinear l_1 -approximation,” *SIAM Journal on Numerical Analysis*, 16, 70–86.
- Gorse, C., Johnston, D., and Pritchard, M. (2012), *A Dictionary of Construction, Surveying, and Civil Engineering*, Oxford: Oxford University Press.
- Harrison, D. and Rubinfeld, D. L. (1978), “Hedonic housing prices and the demand for clean air,” *Journal of Environmental Economics and Management*, 5, 81–102.
- He, X. and Shi, P. (1994), “Convergence rate of B-spline estimators of nonparametric conditional quantile functions,” *Journal of Nonparametric Statistics*, 3, 299–308.
- Milnor, J. W. (1965), *Topology from the Differentiable Viewpoint*, Charlottesville, VA.: The University Press of Virginia.
- Padilla, O. H. M., Tansey, W., and Chen, Y. (2020), “Quantile regression with ReLU Networks: Estimators and minimax rates,” *arXiv preprint arXiv:2010.08236*.
- Schmidt-Hieber, J. (2020), “Nonparametric regression using deep neural networks with ReLU activation function,” *The Annals of Statistics*, 48, 1875–1897.
- Shen, G., Jiao, Y., Lin, Y., Horowitz, J. L., and Huang, J. (2021), “Deep Quantile Regression: Mitigating the Curse of Dimensionality Through Composition,” *arXiv preprint arXiv:2107.04907*.
- Thaler, R. (1978), “A note on the value of crime control: evidence from the property market,” *Journal of Urban Economics*, 5, 137–145.
- Van der Vaart, A. W. and Wellner, J. A. (1996), *Weak Convergence and Empirical Processes*, New York: Springer.

Yeh, I.-C. (1998), “Modeling of strength of high-performance concrete using artificial neural networks,” *Cement and Concrete Research*, 28, 1797–1808.